



DOES THE MODEL KNOW IT'S HALLUCINATING?

CSC3043S – Reasoning in AI: Assignment 2

Abstract

The report details the result from the implemented POPE Benchmark with only Random
Negatives and Adversarial Negatives

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Overview

The model can hallucinate, saying an object is present in an image while the object is not there. It's hallucination, however, is not that interesting. What is interesting is how it happens. The model's final answer and confidence don't tell us much about how it happened. This is because the model first performs a bunch of internal calculations before its final answer and confidence.

The core research question here whether a model's internal representations already contain evidence that its own answer is wrong. We probe the model with simple binary questions to achieve this. The questions are structured as "Is there a {object} in this image? Answer yes or no.". This is because ground truth is ambiguous and we want to avoid/reduce such ambiguity.

We are mostly interested in the adversarial negatives, where the model answers yes for the question even though the object is not there but the object frequently appears with objects that are there. This is the most interesting hallucination because the model might be using common reasoning instead of looking at the actual pixels.

The assignment treats a pretrained Visual Language Model (VLM) as an object of study, and constructs a controlled question set about real images with known ground truth. We use the val2017 dataset. Chose the recommended SmolVLM-256-Instruct as the object of study.

The assignment makes use of a COCOSubset class which wraps a coco object with a torch dataset. To avoid unnecessary computations, the image ids are loaded once and stored, the mapping from category id to category name are computed and stored, so as the image ids to categories mapping.

The assignment defines the need for a symmetric pairwise category co-occurrence counts across every image in the dataset. To avoid have multiple sources for one truth, the co-occurrence category pairs are stored canonically, e.g. ('cat', 'dog') and not ('dog', 'cat'). The lookups are symmetric, as we check for both orders, e.g. ('cat', 'dog') and ('dog', 'cat').

Adversarial categories are chosen together with present categories as a pair with the highest cooccurrence count across the whole dataset. If no such pair was found, the fallback is to choose an adversarial category randomly, which defeats the purpose. However, after running a couple of tests, it is safe to say that the fallback is never triggered, with the fallback rate at 0.00% ([see logs](#)).

PART – A

Question 1

Question Type	QA Pairs Counts
<i>Present</i>	200
<i>Random Negatives</i>	200
<i>Adversarial Negatives</i>	200
Total	600

Using my student number to create a seed, I have seed = 1408026872311599607857. Using this seed to randomly sample image ids and to sample the random negatives, the resulting QA pairs per question type are balanced.

Question 2

The adversarial category for an image is chosen together with the present category as the (present category, absent category) pair with the highest cooccurrence count across the whole dataset (using the precomputed cooccurrence Counter).

For example, for image with image_id=369422, there is a person in the image (present), who mostly appears together with a chair which is currently not present in the image (adversarial category), and an airplane is currently not present in this image (chosen randomly).

PART – B

Question 3

The strict parser marked 0% of generations as unclear because they were no unclear responses. I applied a conservative fallback parser logic:

- Responses containing an unambiguous standalone ‘yes’ or ‘no’ were recovered, while responses with neither or conflicting signals remained unresolved.
- Only the remaining unresolved cases were excluded from accuracy calculations.

Question 4

The overall answer accuracy is about 89.83%. The present questions with 78% accuracy, the absent randoms with 96.5% and the absent adversarial at 95% (see [log](#)). The present question is harder for the model, which is quite unexpected as that was thought to be the absent adversarial, which the model is surprisingly doing very well at.

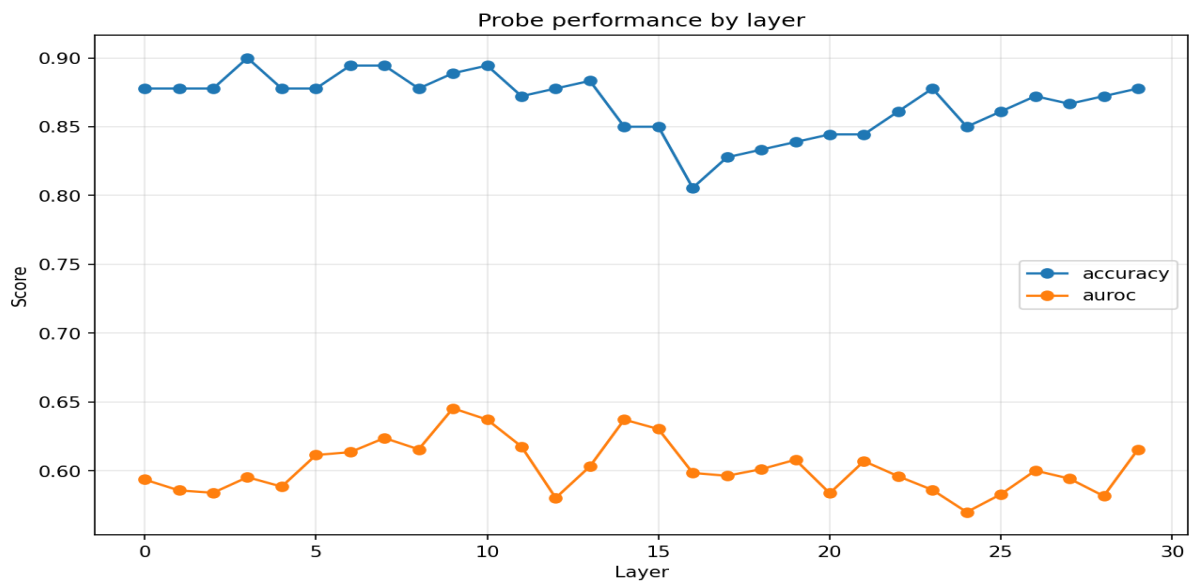
Question 5

I choose mean pooling because model generation is autoregressive. The model might have started hallucinating at the beginning and continued with the hallucination throughout. If we look only at the last token layer, we assume that is where the hallucination usually occurs.

PART – C

Question 6

The probe accuracy and AUROC for every layer is available under [data/layer_auroc.csv](#). From the results and the plot below, we can see that the early layers are more informative about correctness. Accuracy is high (peaking at 90%), while the AUROC score is roughly around 58%. The accuracy might stem from the fact that there are many correct examples and a few hallucinated examples (see [probing.log](#)). The AUROC score tells us that the scores provide a useful signal, but not very high scores are received by correct examples.



Question 7

Method	Accuracy	AUROC
Best-layer probe (layer 9)	0.888889	0.645405
Confidence-only baseline	0.911111	0.904321

With the threshold for correctness set at 65%, the confidence only baseline reports the accuracy and AUROC scores as above. The model performs better than the best-layer probe (at layer 9) by about 25.89% in terms of AUROC score.

PART – D

Question 8

Setting	Accuracy	AUROC
Within-distribution	0.888889	0.645405
Cross-category	0.900000	0.683036

The cross-category did not lose performance: accuracy increased by about 1.1% and AUROC increased by about 3.8% (see [generalization.log](#)). This suggests that the hidden states may encode a signal that generalises beyond the specific construction of present and random-negative examples, rather than only recognising those question types. However, the moderate AUROC of about 68% provides limited evidence for hallucination signal.

Question 9

Example	Image ID	Category	Question type	Ground truth	Model confidence	Probe prediction
1	303713	Car	Absent adversarial	False	0.6865	Hallucinated
2	402720	Chair	Absent adversarial	False	0.5205	Correct
3	103548	Chair	Absent adversarial	False	0.9976	Hallucinated

The baseline's confidence with a threshold of 0.65 gives a better signal of whether the model has hallucinated or not. This is supported by the results in [data/disagreements.csv](#) where the probe was able to detect the hallucination only once given the disagreements.

Looking at the images in the Appendix, you can see that most of the images in question had a person present. Since a person occurs mostly frequently with a chair and a car, the model might sometimes be lazy and use reasoning to make predictions.

Question 10

The moderate AUROC score of about 68% and 64% suggests that the model's hidden states have some information about the hallucination before it is spoken, but it's model's confidence usually shows.

When we used the baseline to detect hallucination and set the model's correctness at greater than 65% of it's confidence, the model was disagreed with the probe only once, where its confidence was at 68%. Had a higher threshold of about 75% or 70% been used, baseline will have been able to detect when the model is hallucination. Looking at the result's in [row_metadata.json](#), we can see that the model is almost always right when its confidence is level is very high. This is also backed by the AUROC score of the baseline in PART C of about 90%.

So, yes, the model's hidden states do carry some signal of its hallucination, however, a larger sample size could better expose this. Additionally, for PART D, we could have trained the probe on adversarial examples of the training data and then evaluated on the adversarial examples of the validation data. This could have exposed the model's hidden states the secret even better.

However, looking at the results in [logs/probing.log](#) under sanity check, we can see that model rarely hallucinates the random absent and the absent adversarial categories, but do hallucinate the present ones slightly more often.

AI Declaration

I have used generative AI to help breakdown the assignment requirements and explain in clear terms what is required of me. I have made use of generative AI to help freeze the SmolVLM and how to configure it and use it. It explained what the hugging face API exposes to me and how I can use it. I have used generative AI to help debug my code and give suggestions of potential improvements to the code. All work done by AI have been declared.

Appendix



Figure 1: image_id=303713



Figure 2: image_id=402720



Figure 3: image_id=103548